ELECTRONIC AND AI-GENERATED EVIDENCE IN JUDICIAL PROCEEDINGS

Authenticity, Integrity, Deepfakes, Chain of Custody and the Right to an Effective Challenge

PROFESSIONAL SCHOLARLY PAPER

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DOI: 10.5281/zenodo.23017531

AI Advokat · Skopje · 2026

Professional and scholarly analysis with a practical evidential protocol for lawyers, courts, prosecution services and expert witnesses

This paper does not replace legal or forensic assessment in an individual case

Abstract

This paper examines electronic and AI-generated evidence as a distinct evidential phenomenon that is reshaping the factual, technical and procedural foundations of judicial decision-making. Its central thesis is that electronic form does not, in itself, diminish evidential value, but it does require verifiable safeguards concerning provenance, integrity, context and the manner of acquisition. With AI-generated and synthetic content, traditional visual or auditory persuasiveness is no longer sufficient. Courts must distinguish admissibility, authenticity, integrity, reliability and evidential weight, while a party must have a genuine opportunity to examine the methodology, data, tool version, error rate and chain of custody. The analysis links Macedonian procedural rules with Articles 6 and 8 of the European Convention on Human Rights, the Budapest Convention, its Second Additional Protocol, the European e-Evidence package, the AI Act and contemporary digital-forensic standards. The practical contribution consists of a five-stage test for judicial assessment, a chain-of-custody model, a protocol for suspected deepfakes, questions for expert witnesses and checklists for lawyers and courts.

Keywords: electronic evidence, AI-generated evidence, deepfake, authenticity, integrity, metadata, chain of custody, expert evidence, fair trial, effective challenge

Scholarly and Professional Contribution

The original contribution of this paper is the systematisation of electronic evidence through five distinct questions: lawfulness, authenticity, integrity, reliability and the possibility of effective challenge. The model avoids the mistaken binary approach under which electronic evidence is treated as either entirely reliable or entirely worthless. Instead, each weakness is linked to a specific procedural consequence: exclusion, additional expert examination, reduced evidential weight or the need for corroborating evidence.

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1 Introduction

Electronic evidence is no longer exceptional. Email, messages from communications platforms, photographs, video, GPS records, logs, cloud data, electronic signatures and automated reports are routinely used in civil, criminal, administrative, employment and commercial cases. The new phase is the appearance of content that is not merely stored electronically, but has been created or materially altered by AI.

The practical problem is not simply whether electronic content can constitute evidence. A copy may appear identical to the original; an alteration may be invisible; metadata may be lost when material is forwarded; and synthetic audio or video may persuasively depict an event that never occurred. At the same time, an allegation that something is a deepfake may itself be misused to discredit authentic evidence.

Modern evidence law must therefore navigate between two dangers: the uncritical acceptance of technologically persuasive content and the automatic rejection of electronic evidence merely because manipulation is possible. The proper approach is to examine provenance, integrity, context and the possibility of adversarial analysis.

1.1 Central Thesis

The central thesis is that the evidential value of electronic content does not derive from how convincing it looks, but from a verifiable evidential pathway. The greater the possibility of manipulation, and the more significant the fact to be proved, the stronger the safeguards should be concerning the original, metadata, forensic copy, hash value, documentation of access and independent expert examination.

1.2 Terminological Note

In this paper, electronic evidence means any information of potential evidential significance that is created, processed, transmitted or stored in digital form. AI-generated evidence means content or an analytical output created wholly or partly by AI. A deepfake is a realistic synthetic or manipulated image, audio or video that may falsely appear authentic. These terms do not determine evidential value; they merely define the object of examination.

2 Subject, Methodology and Hypotheses

2.1 Subject and Purpose

The subject is the judicial use and challenge of electronic and AI-generated evidence. The purpose is to create an integrated legal and technical model applicable to lawyers, judges, prosecutors and expert witnesses, without uncritically transplanting standards from any single jurisdiction.

2.2 Methodology

Normative, functional, comparative and case-study methods are used. The normative method examines procedural rules, human rights, data protection and international co-operation. The functional method follows the evidence from creation to presentation in court. Comparative sources are used as persuasive standards rather than as automatically applicable domestic law. The hypothetical case tests the proposed model within adversarial proceedings.

Method	Key question	Limitation
Normative	What conditions follow from the law in force?	There is no single code governing electronic evidence
Functional	What happened to the data throughout its life cycle?	The technical trail depends upon the system
Comparative	Which standards are developing in Europe?	Foreign rules are not domestic sources of law
Case study	Does the proposed test work in a dispute?	The facts are hypothetical

2.3 Research Questions

- When is electronic content sufficiently authenticated for use in proceedings?
- What is the difference between an original, a forensic copy, a screenshot and a forwarded file?
- Which weaknesses in the chain of custody affect admissibility, and which affect weight?
- How should the burden be allocated where one party alleges that material is a deepfake?
- What information must be available for an effective challenge to an algorithmic result?

2.4 Working Hypotheses

H1: electronic form does not automatically require a higher or lower standard of proof, but rather a form of scrutiny adapted to risk. H2: a hash value is important evidence that a particular file has remained unchanged, but it does not prove that the content was true when created. H3: suspicion of a deepfake is not, without a concrete basis, sufficient for exclusion, but may justify preservation of the original and expert examination. H4: the right to an effective challenge requires access to sufficient information about provenance, method and error, subject to proportionate protection of legitimate secrets.

3 Concept and Classification

3.1 Classification by Origin

Category	Example	Principal risk
Native digital evidence	Email, log, GPS record	Identification of user and system
Digitised copy	Scanned contract, photograph of a document	Absence of original and context
User communication	Viber, WhatsApp, SMS	Identity, completeness and export method
Machine-generated record	Server log, telemetry	Clock, configuration and automation
AI-generated content	Text, image, audio, video	Synthetic provenance and manipulation
AI analytical result	Score, classification, recommendation	Method, data and error rate

3.2 Five Distinct Evidential Questions

Admissibility asks whether the evidence may procedurally be used. Authenticity asks whether it is what the party says it is. Integrity asks whether it has been altered since the relevant point in time. Reliability asks whether the content and method deserve confidence. Evidential weight asks how persuasively the item proves the disputed fact in the context of the case as a whole.

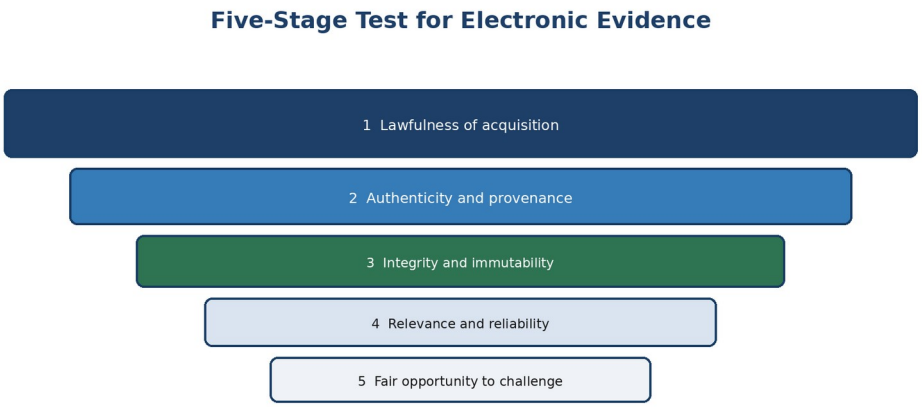


Figure 1 Five-stage test for judicial assessment of electronic evidence

3.3 The Original in a Digital Environment

With digital evidence, an original does not always mean a single physical carrier. Several bit-for-bit identical copies may exist. The practical question is therefore whether the file and relevant metadata have been preserved in a manner that permits verification. A screenshot is a visual reproduction and may be useful, but it usually does not contain all headers, metadata, attachments and system information.

4 Normative Architecture

4.1 Domestic Framework

The Macedonian framework is built from procedural statutes, rules on expert evidence, the Law on Electronic Documents, Electronic Identification and Trust Services, legislation governing electronic communications, personal-data protection and the substantive law relevant to the dispute. It is not enough that evidence technically exists; it must have been obtained and used in a manner that respects privacy, confidentiality of communications and the rights of the defence.

In criminal proceedings, Articles 181 and 184 of the Law on Criminal Procedure are particularly relevant to judicial authorisation and the search of computer systems and data, while Articles 194–198 govern the seizure of items and computer data. These provisions do not create a separate “code” of electronic evidence, but they establish the procedural framework for lawful access, scope and preservation.

4.2 European Convention on Human Rights

Article 6 does not establish a detailed European code on the admissibility of evidence. The European Court of Human Rights primarily asks whether the proceedings as a whole were fair, taking account of the manner in which evidence was obtained, the opportunity to challenge it, its quality and its importance to the outcome. Where evidence has been unlawfully obtained, Article 8 and the particular guarantees against ill-treatment may also be relevant. A court should not conceal the weakness of a crucial item of electronic evidence behind a general formula stating that the evidence has been assessed individually and as a whole.

Of particular relevance to electronic evidence is *Yüksel Yalçınkaya v. Türkiye* [GC], no. 15669/20 (2023), in which the Court emphasised the need for an effective opportunity to challenge key digital evidence and its technical basis. The general framework concerning unlawfully obtained evidence is also reflected in *Schenk v. Switzerland*, *Bykov v. Russia* [GC] and *Gäfgen v. Germany* [GC]. In EU law, C-670/22 (EncroChat, 2024) is relevant to the cross-border acquisition and use of electronic evidence.

4.3 Budapest Convention

The Budapest Convention provides a framework for procedural powers and international co-operation concerning computer data and electronic evidence. The Second Additional Protocol, opened for signature in 2022, develops mechanisms for enhanced co-operation and disclosure of electronic evidence, including direct co-operation with service providers in defined circumstances, subject to human-rights and data-protection safeguards. The Republic of North Macedonia signed the Second Additional Protocol (CETS No. 224) on 12 May 2022; according to the Treaty Office status in September 2026, no ratification was recorded.

4.4 European e-Evidence Package

Regulation (EU) 2023/1543 establishes European Production Orders and European Preservation Orders for electronic evidence in criminal proceedings and has applied within the EU since 18 August 2026. Together with Directive (EU) 2023/1544, it creates a new infrastructure for the cross-border acquisition of data from service providers. For the Republic of North Macedonia, the regime is not directly applicable domestic law, but it is relevant as a European regulatory development, particularly in cases involving providers or data located in the EU.

4.5 The AI Act and Deepfake Transparency

The AI Act establishes transparency rules for certain synthetic and manipulated content. Regulation (EU) 2026/1744 amended the transitional framework: for certain systems placed on the market before 2 August 2026, the obligation concerning machine-readable marking under Article 50 applies after the additional transitional period ending on 2 December 2026. Labelling may assist, but its presence or absence does not determine judicial authenticity, lawfulness or evidential weight.

5 Admissibility and Lawfulness

5.1 Lawfulness Is Not the Same as Truth

Evidence may be technically authentic but unlawfully obtained; or lawfully obtained but unreliable. The court must examine separately the legal basis for obtaining it, the authority of the body or party, the scope, necessity, proportionality and the manner of storage.

5.2 Privately Obtained Evidence

Where evidence has been obtained by a private party, the court should examine ownership of the device, authority to access it, the reasonable expectation of privacy, the method of acquisition and any possible breach of the confidentiality of communications. The fact that evidence is relevant does not automatically cure unlawfulness. The procedural consequence, however, depends upon the type of proceedings, the seriousness of the infringement and the fairness of the proceedings as a whole.

5.3 Video Surveillance

Video-surveillance footage raises two groups of questions: whether the surveillance itself was lawful and whether the particular recording is authentic and complete. Domestic personal-data rules require a defined purpose, transparency, restricted access, technical and organisational measures and regulation of retention periods. These factors may be relevant both to lawfulness and to evidential reliability.

5.4 Proportionate Procedural Response

Defect	Possible step	Ultimate measure
Unclear legal basis	Additional reasons and disclosure	Exclusion where the violation requires it
Excessively broad scope	Redaction and restricted use	Exclusion of irrelevant material
Breach of privacy	Necessity and fairness assessment	Separate remedy or exclusion
Insufficient technical trail	Expert evidence and corroboration	Reduced weight or non-admission

6 Authenticity

6.1 What Must Be Proved

Authenticity does not mean that the content is true. It means that the evidence originates from the asserted source or person and is what the proponent claims it to be. An email may genuinely originate from a particular account, while the account itself was compromised. A video may genuinely have been created on a particular phone while depicting a staged event.

6.2 Methods of Authentication

- testimony from a person who created, received or exported the content;
- system logs, server headers and provider data;
- electronic signature, electronic seal or time stamp;
- forensic copy and hash values;
- characteristics known only to the participants;
- continuity with other independent evidence;
- expert examination of the device, file and medium.

6.3 Screenshots and Printed Copies

A screenshot may show what was visible on a screen at a particular moment, but it can easily be edited and often omits the broader conversation, identifiers, time zone and technical headers. Where the evidence is disputed, the conversation export, device, backup, provider data and testimony about how the content was created should be sought. A printed copy should not be treated as the complete equivalent of the source file.

6.4 Electronic Signatures and Trust Services

A qualified electronic signature may provide a strong presumption of identity and integrity within the applicable legal framework, but the court should still establish precisely what was signed, the validity of the certificate, the relevant time and any possible compromise of the signature-creation device.

7 Integrity and Metadata

7.1 Integrity

Integrity concerns whether the evidence has remained unchanged from the relevant point in time until presentation. It is demonstrated through a combination of hash values, write-blocking, controlled forensic copying, logs, restricted access, documentation of each opening and comparison with the source. A bare assertion that a file was held in an official system is insufficient if no information is available about access and alterations.

7.2 Hash Values

A cryptographic hash is a digital fingerprint of the bit-level content. A matching hash shows, with a high degree of confidence, that the files compared are bit-for-bit identical. A hash does not prove who created the file, whether the scene depicted is true, or whether manipulation occurred before the first hash was calculated. It is therefore one part of the evidential pathway, not an independent certificate of truth. For new forensic processes, a contemporary algorithm such as SHA-256 or stronger should be used. MD5 and SHA-1 should not be relied upon as the sole guarantee of integrity because of known collision weaknesses.

7.3 Metadata

Type	Examples	Risk
System	Creation, modification, user, path	May change through copying or configuration
Communications	Sender, recipient, server, Message-ID	Spoofing or compromised account
Media	EXIF, device, location, orientation	Deletion or manual alteration
Cloud	Version, access, download, audit log	Provider dependence and retention period
Application	Message ID, delivered/read status	Differences by version and export method

7.4 Time Zones and System Clocks

Differences between local time, UTC, server time and an incorrectly configured clock may alter the chronology. The expert should state which time zone was used, whether the system synchronised its clock and how any discrepancies were corrected.

8 Chain of Custody

The chain of custody is the documented history of the evidence: who identified it, when and how it was acquired, with which tool, where it was stored, who had access, whether a forensic copy was created and whether the hash values match. Its purpose is to permit reconstruction, not to encourage mechanical completion of a form.

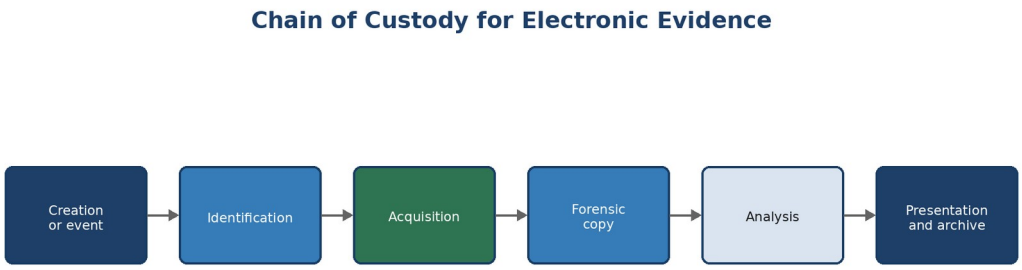


Figure 2 Model chain of custody for electronic evidence

8.1 Minimum Fields

Field	Content
Identifier	Unique evidence and case number
Source	Device, profile, service or person
Date and time	With time zone specified
Method of acquisition	Tool, version, commands and parameters
Hash	Algorithm and value for source and copy
Custodian	Person, location and access rights
Transfer	From whom to whom, reason and signature
Change	Every processing step, conversion or redaction

8.2 Breaks in the Chain

Not every administrative omission renders evidence worthless. The court should determine whether the defect creates a genuine possibility of substitution, alteration or mistaken linkage, and whether the weakness can be overcome by other evidence. Particularly serious are unexplained access to the original, a hash mismatch, loss of the source device or analysis performed on a mutable working copy without a preserved original.

9 Deepfakes and Synthetic Content

9.1 The Dual Risk

Deepfakes create a dual risk. The first is that false content may be accepted as genuine. The second is that authentic content may be dismissed as false merely because modern manipulation is possible. Neither visual persuasiveness nor bare denial is therefore sufficient.

9.2 Provenance, Content Credentials and the Limits of Technical Provenance

Modern provenance mechanisms, including cryptographically signed manifests and Content Credentials, can provide verifiable information about the creation and alteration of media. Their evidential function is strongest when they demonstrate continuity of origin, but they are not a substitute for judicial authentication. A valid provenance record may confirm which system or device processed a file without proving that the event shown actually occurred; conversely, the absence of provenance does not mean

that content is false. Provenance should therefore be treated as one layer in the evidential pathway, alongside the source, metadata, chain of custody, testimony and forensic analysis.

9.3 Indicators Are Not Conclusions

Artefacts in a face, lighting, lip movement, audio or compression may justify investigation, but they are not conclusive evidence. Platforms recompress content, and ordinary editing may create similar traces. Automated deepfake detectors also have error rates, dataset dependencies and the potential to become outdated.

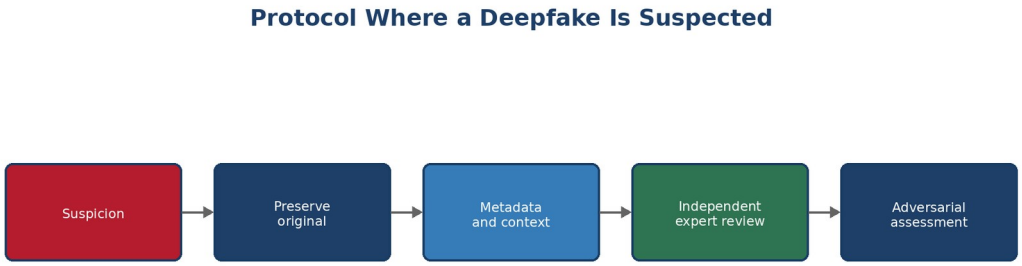


Figure 3 Protocol where a deepfake is suspected

9.4 Protocol

1. Secure the source file and prevent automatic deletion.
2. Document the channel through which it was received and each prior conversion.
3. Calculate a hash and create a forensic copy.
4. Collect metadata, cloud logs and corroborating records.
5. Instruct an independent expert using a validated methodology.
6. Give the opposing party access to sufficient material for counter-analysis.
7. Require the court to explain the methodology, limitations and evidential weight.

9.5 Burden Where Manipulation Is Alleged

The party adducing the evidence should provide a sufficient initial basis for authenticity. A party alleging manipulation should identify specific reasons where reasonably possible. Where, however, crucial technical material lies exclusively under the control of the proponent or a third party, the court should avoid imposing an impossible burden of proof and should order preservation, disclosure or expert examination as appropriate.

10 AI-Generated Text, Images, Audio and Video

10.1 AI-Generated Text

Text may be evidence of what someone sent regardless of whether it was composed by a human or by AI. If, however, it is offered as evidence of knowledge, intention or authorship, the use of AI may be material. Versions, changes, prompts, profile information, logs and human interventions may be required where available and proportionate.

10.2 Images and Video

With a generated or altered image, it is necessary to distinguish among wholly synthetic media, AI enhancement, object removal, background alteration and ordinary correction. Each intervention has a different significance. Enhanced brightness may be permissible demonstrative processing if the original is preserved and the method disclosed; adding a person alters the factual content.

10.3 Cloned Voice

Similarity of voice is not enough for identification. The source recording, transmission channel, codec, spectral characteristics, context, originating number and independent facts should be examined. A short and compressed recording may limit the reliability of any expert conclusion.

10.4 AI Analytical Output

Where the evidence consists of a score, classification or automated recommendation, the output alone is not the object of scrutiny. The purpose, version, input data, relevant parameters, validation, error rate, applicability to the particular population, logs and human interventions are also required. The explanation must be specific to the case rather than a marketing description of the model.

11 Evidence from Communications Platforms

11.1 Viber, WhatsApp, Messenger and SMS

Objections most often concern the identity of the profile, completeness of the conversation, deleted messages, altered contact names and screenshots lacking a source. The strongest combination is the source device, logical or physical extraction, an export with attachments, backup, data from the operator or service where available, and testimony from a participant.

11.2 Context and Selective Extracts

A single message may carry the opposite meaning when read together with the messages before and after it. The court should seek reasonably complete context while avoiding unnecessary disclosure of irrelevant private communications. Redactions should be documented, and the complete version may be placed under a protective regime.

11.3 Email

A printed email does not reveal the complete headers, the route through servers, Message-ID or SPF, DKIM and DMARC results. Those elements may assist, but do not exclude the possibility of a compromised account. Server logs, retention policies and testimony concerning ordinary business practice may also be important.

Evidence	Minimum package	Additional verification if disputed
App message	Screenshot plus identification of device and profile	Export, backup, forensic extraction
Email	Complete message with headers	Server logs, DKIM and administrator evidence
SMS	Device, number and complete conversation	Operator data and extraction
Social network	URL, profile, date and preserved page	Platform data and web archive

12 Cloud and Cross-Border Evidence

12.1 Location and Control

Cloud evidence may physically exist in several states while legal control is exercised by a provider in a third state. Questions of jurisdiction, authorisation, voluntary disclosure, data protection and professional secrecy must be analysed separately from the technical ability to download the data.

12.2 Urgent Preservation

Cloud logs and user data may be deleted after a short retention period. The first step is often not a request for full disclosure but a precise preservation order or request. It should identify the account, period, categories of data and time zone, without disproportionate collection.

12.3 International Co-operation

The Second Additional Protocol and the e-Evidence regime accelerate co-operation with service providers, but do not remove the need for lawfulness, speciality, data protection, remedies and procedural scrutiny. The provider becomes an important custodian of evidence, but its certificate should not automatically replace the opportunity to challenge.

13 Digital Forensics and Expert Evidence

13.1 Mandate of the Expert

The expert should answer technical, not legal, questions. The expert may establish hashes, metadata, traces of alteration, configuration, error rates and reproducibility. The expert should not decide whether evidence is legally admissible, whether an accused person is guilty or which witness is credible.

13.2 Minimum Content of the Expert Report

Element	What it must contain
Subject matter	Precise identification of devices, files and copies
Method	Tools, versions, parameters and procedure
Integrity	Hash before and after analysis and access history
Result	Separation of observation from interpretation
Validation	Tests, reference material and error rate
Limitations	Missing data, compression and uncertainty
Reproduction	Sufficient steps for independent verification

13.3 Tools and Versions

Relevant technical reference frameworks include ISO/IEC 27037 (identification, collection and preservation of digital evidence), ISO/IEC 27041 and 27042 (validation of methods and analysis), NIST SP 800-101 Rev.1 for mobile-device forensics, SWGDE best practices, ENFSI guidance on forensic evaluation and C2PA Content Credentials for the provenance of media content. None of these standards, by itself, determines evidential admissibility; they strengthen technical verifiability.

The name of a well-known forensic tool is not a substitute for methodology. The version, module, settings, licence, testing and possible errors must be stated. Where the tool uses AI, the dataset, conditions of use and stability of the output should also be examined.

13.4 Questions for the Expert

- Was the analysis performed on the original or on a forensic copy?
- Which hash values were calculated, and when?
- What changes does the tool make when opening or converting the material?
- Has the method been validated for this type of device, format, language and compression?
- What is the known or estimated false-positive and false-negative rate?
- Can another expert reproduce the result using the material supplied?
- What alternative explanations remain possible?

14 Right to an Effective Challenge

14.1 A Practical, Not Merely Formal, Opportunity

The right to challenge is not satisfied merely because a party can verbally deny the evidence. The party must receive, in sufficient time, enough information and access to the material to instruct an expert, propose a counter-test and examine the persons who obtained and analysed the evidence. This is especially important where electronic evidence is the sole or decisive evidence.

14.2 Trade Secrets and Security

A provider or proponent may invoke trade secrecy, security or the privacy of third parties. The solution need not be complete public disclosure. Protective inspection, redacted documentation, a confidential expert, restrictions on copying and judicial review are all possible. The measure must nevertheless preserve a sufficient core for an effective defence.

14.3 Adverse Procedural Inferences

Where the party controlling crucial technical material fails to preserve it or unjustifiably refuses to disclose it, the court may take that circumstance into account when considering the burden and evidential weight, in accordance with the applicable procedural law. Such an inference must not operate as an automatic sanction; control, the duty to preserve, fault and the importance of the lost material should first be established.

15 Burden and Standard of Proof

15.1 Initial Authentication

The proponent bears the initial burden of showing what the evidence is and where it came from. The intensity required depends upon the nature of the evidence, the objection and its importance. For an undisputed administrative email, testimony and the complete message may suffice. For disputed video that alone proves a criminal offence, a complete forensic trail may be required.

15.2 Particularised Objection

Where possible, an objection should identify the weakness: an unknown device, a break in custody, time discrepancy, lost metadata, compromised account, editing traces or an unverifiable AI detector. A bare assertion that anything can be forged should not be enough.

15.3 Information Asymmetry

Where the information is held exclusively by a public authority, provider or opposing party, placing a strict burden upon the challenger may render the right illusory. The court should use procedural orders, preservation measures, expert evidence and an appropriate allocation of the burden.

16 Application in Civil, Criminal and Administrative Proceedings

16.1 Civil Proceedings

In civil proceedings, electronic evidence is often filed as documents, photographs or transcripts. Effective case management requires early identification of disputed authenticity, timely requests for the original and metadata, preservation of evidence and a precisely defined task for the expert. Procedural preclusion should not be applied mechanically where the need for forensic examination only became apparent after the evidence was challenged.

16.2 Criminal Proceedings

Criminal proceedings require particular rigour because of the presumption of innocence, the lawfulness of searches and seizure, the right to silence, access to material and the opportunity for counter-expertise. The prosecution should not rely upon an analytical report where the underlying data are unavailable and the methodology unexplained.

16.3 Administrative Proceedings and Administrative Disputes

Where an authority relies upon logs, automated scores or electronic records, the reasons must show which data were used, where they came from, how the party can correct them and what human review was carried out. A system extract should not become irrebuttable evidence merely because it was produced by a public authority.

16.4 Employment and Family Cases

Employment cases may involve email, access logs, GPS data and video surveillance, with substantial risks to privacy and proportionality. Family cases may involve private messages and recordings containing sensitive data relating to children and third parties. The court should confine the scope to material that is genuinely relevant.

17 Evidential Protocol for Lawyers

17.1 First Steps

1. Identify the source and the person who controls it.
2. Send a preservation request before the retention period expires.
3. Do not open or alter the original without a plan.
4. Create a forensic copy using documented tools and a hash.
5. Prepare a chronology of all access and transfers.
6. Separate legal objections from technical objections.

17.2 Decision Matrix

Question	If yes	If no
Is the evidence crucial to the outcome?	Enhanced scrutiny and an independent expert	Proportionate scrutiny
Is the original available?	Forensic copy and hash	Explain the loss and seek other sources
Are the metadata preserved?	Compare them with the content and logs	Do not presume integrity
Is there a specific deepfake concern?	Preserve and test by multiple methods	Avoid unnecessary expensive examination
Does the opposing party lack access?	Provide adversarial inspection	Document equal conditions

17.3 Application to Preserve Evidence

The request should be precise in time, subject matter and technical scope: it should seek preservation of the source device or profile, original file, relevant metadata, audit logs, versions, backup, reports and documentation of each conversion. It should explain the risk of automatic deletion and propose a protective regime for irrelevant or confidential data.

17.4 Preparation for the Hearing

- prepare an evidential map linking fact, source, objection and response;
- identify which witness can explain creation and custody;
- prepare technical questions that avoid legal conclusions;
- ensure appropriate equipment for presentation and an identical copy for the parties;
- ask the court to rule separately on lawfulness, authenticity, integrity and weight.

18 Hypothetical Case

18.1 Factual Setting

The company ORION seeks MKD 9,400,000 in damages from a former managing director for the alleged diversion of clients. The key evidence is an audio recording submitted anonymously through a cloud link, on which a voice resembling the former manager appears to negotiate a commission. The claimant also produces a screenshot of a WhatsApp conversation. The defendant contends that the audio contains a cloned voice, that the conversation was assembled from several messages and that the original telephone has not been produced.

18.2 Claimant's Theory

The voice, content and business details identify the defendant. Two employees recognise the voice. The screenshot contains messages dated consistently with the loss of the client. The claimant's expert found no obvious breaks or artefacts.

18.3 Defendant's Theory

The recording has no known source, source device, original metadata or chain of custody. The cloud file is compressed. The claimant's expert method detects only crude editing, not voice cloning. A screenshot can be altered, a contact name does not prove the number, and the messages before and after the disputed passage are missing.

18.4 Evidential Matrix

Issue	Claimant's evidence	Objection	Required measure
Voice identity	Recognition by employees	Subjectivity and voice cloning	Independent forensics and comparison samples
Audio provenance	Cloud link	Unknown device and creator	Request logs and upload history
Integrity	No visible breaks	No original and no hash from creation	Reduced weight and corroboration
WhatsApp conversation	Screenshot	Selection and editable contact name	Device, export, backup and number
Loss	Lost client	No causation	Contracts, communications and financial evidence

18.5 Order for Preservation of Evidence

The court orders the claimant to produce the earliest preserved version of the audio, cloud audit logs and all communications in its possession with the anonymous source. An independent expert is appointed and a protective regime established for the voice-comparison samples. As regards WhatsApp, the source device or a verifiable export is required. If the claimant cannot produce it, the screenshot is not automatically excluded, but its evidential weight is materially reduced.

18.6 Questions for the Expert

- What is the earliest available generation of the recording, and how many times has it been converted?
- Can the method distinguish voice cloning from compression and noise reduction?
- Which models and synthesis techniques are covered by the test?
- What are the error rates for a recording of this length and quality?
- Are the comparison samples temporally and technically appropriate?
- Can authenticity be established, or only the absence of detected traces of manipulation?

18.7 Hypothetical Judgment

The court finds that the audio has not been sufficiently authenticated to prove the disputed agreement on its own. The unknown provenance, lost metadata and limitations of the analysis prevent a reliable conclusion that the recording is authentic, but they do not prove that it is a deepfake. The screenshot is given limited weight because the source device and context are missing. The claim is assessed on the basis of the remaining contractual, witness and financial evidence. The court expressly avoids adopting a rule that every suspicious digital medium is inadmissible; it decides according to the cumulative quality of the evidential pathway.

18.8 Grounds of Appeal

Ground of appeal	Response
The court unlawfully required proof of absolute authenticity	Absolute certainty was not required; a sufficient initial basis was required for crucial evidence
The court should have excluded the audio as a deepfake	A deepfake was not proved; the weakness affected weight
Voice-recognition testimony was sufficient	Recognition was relevant but did not remove the technical weaknesses
The missing telephone should not count against the claimant	The consequence depends upon control, duty and the opportunity to preserve

19 Recommendations for the Republic of North Macedonia

19.1 Legislative and Institutional Recommendations

1. Develop a technology-neutral definition of electronic evidence and clear rules for preservation and forensic copying.
2. Align cross-border evidence processes with the Budapest Convention and European e-Evidence developments.
3. Establish minimum standards for laboratories, tools, experts and documentation.
4. Provide protective regimes for trade secrets, privacy and counter-expertise.
5. Establish training for judges, prosecutors, lawyers, police and expert witnesses.
6. Develop deepfake protocols without automatically imposing an impossible burden upon the challenger.

19.2 Judicial Practice

Decisions should give separate reasons concerning lawfulness, authenticity, integrity, reliability, weight and the possibility of challenge. A formula stating that evidence has been assessed individually and as a whole is not sufficient where the outcome depends upon disputed electronic evidence.

19.3 The Legal Profession

The Bar Association and professional organisations should develop practical training on collection, preservation, cloud requests, metadata, expert evidence and AI manipulation. Law firms should have an internal procedure for receiving electronic material and a rule that an original must not be processed without a copy and an audit record.

20 Conclusion

Electronic evidence is, by its nature, neither more reliable nor less reliable than traditional evidence. Its distinctive features are that it can be copied easily, altered invisibly, lost quickly and often depends upon systems controlled by third parties. AI further weakens the assumption that realistic appearance is a sign of authenticity.

Responsible judicial assessment requires a cumulative approach: lawful acquisition, verifiable provenance, preserved integrity, appropriate methodology, context, independent verification and a genuine right to challenge. A hash, metadata, a detector or an expert opinion is not a magic solution. Each element has a limited domain and must be assessed alongside the rest of the evidence.

For the lawyer, the central rule is early preservation and precise challenge. For the expert, it is transparent and reproducible methodology. For the court, it is a reasoned decision that exposes the evidential pathway and the fairness of the process. Only then can technological complexity be prevented from becoming a source of unprovable assumptions or unassailable evidence.

Annex 1 Quick Checklist for Lawyers

Control	Yes	No	Note
Source and custodian identified	☐	☐	
Preservation request sent	☐	☐	
Original protected from alteration	☐	☐	
Forensic copy created	☐	☐	
Hash values calculated	☐	☐	
Metadata preserved	☐	☐	
Chain of custody documented	☐	☐	
Lawfulness and privacy assessed	☐	☐	
Expert task defined	☐	☐	
Opportunity for counter-analysis secured	☐	☐	

Annex 2 Chain-of-Custody Form

Date and time	Evidence	Action	From	To	Hash and signature

Annex 3 Court Checklist

Question	Finding
What fact is being proved?	_____
How was the evidence obtained?	_____
What is the original or earliest version?	_____
How were provenance and identity established?	_____
Was integrity preserved?	_____
Which metadata are missing?	_____
What are the method and error rate?	_____
Did the party have sufficient material to challenge?	_____
What role does the evidence play in the final conclusion?	_____

Annex 4 Model Preservation Request

It is requested that, without delay and without alteration or deletion, all data relating to the precisely identified account, device or file for the specified period be preserved, including content where legally permitted, subscriber data, traffic data, audit logs, access records, versions, backup copies, metadata, identifiers, time-zone information and documentation of every automatic or manual alteration. The request does not, by itself, constitute a request for disclosure where a separate order is required for that purpose. The recipient should document the time, scope and method of preservation.

Bibliography

Sources and Verifiability

The legal and institutional sources in this version were editorially verified up to 28 September 2026. European rules are cited with a clear distinction between law directly applicable within the EU and their comparative or approximation relevance for the Republic of North Macedonia. Technical standards are used as professional methodological sources and are not presented as procedural rules.

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Statement of Independence and Limitation

This is an independent professional and scholarly work. The hypothetical case does not concern real persons or proceedings. The technical protocols are models that must be adapted to the particular device, system, proceedings and law in force. The author declares no conflict of interest.